

Henrique Lopes

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Profile

Junior Software Engineer with experience in Python, Web Development and AI integration. Skilled in building interactive applications with Flask, FastAPI, React and Angular, as well as integrating LLMs (GPT, LangChain) and ML frameworks (PyTorch, TensorFlow) into real products. Strong background in research, prototyping and teamwork, motivated to grow in software/AI roles.

Education

Faculty of Engineering, University of Porto (FEUP)	Porto
MSc in Multimedia – Interactive Technologies and Digital Games	2024–2025
University of Minho	Braga
BSc in Engineering Physics	2017–2023

Professional Experience

INESC TEC	Porto
AI Developer Intern	2024–2025
- Built VR applications integrating GPT-based conversational AI with Unity and Python APIs. - Developed functional prototypes using Flask, FastAPI and REST APIs. - Applied ML workflows with PyTorch and TensorFlow for prototyping and testing AI models. - Collaborated with multidisciplinary teams (developers, designers, researchers).	

Selected Projects

- ThinkAlike:** Collaborative platform with AI-assisted UI. (React, Node.js, LangChain)
- StoryShow:** Interactive recommendation platform for movies, series and books, leveraging AI to provide personalized suggestions. (PHP, MySQL)
- Smart Health Tracker:** Health monitoring web/mobile application that collects user data (fitness, sleep, nutrition) and provides personalized ML-powered recommendations. (Python: Flask/FastAPI, React Native, PostgreSQL, TensorFlow)
- Virtual Moderator in VR:** Master’s thesis project in collaboration with INESC TEC: Developed a VR environment for social interaction, integrating GPT-based conversational AI for real-time user engagement. (Unity, Python, GPT, VR)

Technical Skills

- Programming:** JavaScript (ES6+), TypeScript, C, C++, C#, Haskell, Assembly, Python.
- Web Development:** React 18, React Router v6, React Hooks, React Hook Form, Angular, Node.js, Express, Flask, Django, FastAPI, PHP.
- Frontend & UI/UX:** HTML, CSS, Tailwind CSS, CSS Modules, p5.js, Figma, prototyping, accessibility (WCAG).
- Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Redis), Mongoose, SQLAlchemy ORM.
- APIs & Integrations:** REST, GraphQL, Axios, Fetch API, i18n/React i18next, JWT, CORS, Async/await, Promises, HTTP, TLS.
- Artificial Intelligence:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, Chroma, Llama, GPT, Deepseek, AI integration into applications.
- DevOps & Cloud:** Docker, Docker Compose, Kubernetes, AWS (CI/CD), Azure, Ansible, Nginx, Terraform.
- Tools & Monitoring:** Git, GitHub, VS Code, Postman, Swagger, Flasgger, Prometheus, Grafana, Loki, Sentry, ESLint, Jest, Jasmine.
- Collaboration & Management:** Scrum, Kanban, Agile methodologies.

Languages

- Portuguese: Native
- English: Advanced